

Math toolkit for the delegation / coverage argument

Working notes

1 The two empty things

Think of a SPARQL result as a *table*.

Object	Notation	Meaning
Empty solution <i>set</i>	$\emptyset$	no rows: the pattern did not match
Empty solution <i>mapping</i>	$\mu_\emptyset$	one row, zero columns
Join identity	$\{\mu_\emptyset\}$	$\Omega \bowtie \{\mu_\emptyset\} = \Omega$

The cleanest instance is **ASK**, which is exactly what Δ computes:

$$\text{ASK } \{ \langle a \rangle \langle p \rangle \langle b \rangle \} = \begin{cases} \text{true} & \text{iff } \llbracket \cdot \rrbracket_G = \{\mu_\emptyset\}, \\ \text{false} & \text{iff } \llbracket \cdot \rrbracket_G = \emptyset. \end{cases}$$

So $\{\mu_\emptyset\}$ means *yes* and $\emptyset$ means *no*. Note $\{\mu_\emptyset\} \neq \emptyset$: it is a one-element set. Writing $\emptyset$ for “matched with no bindings” would invert the indicator. Notation: $\mu_\emptyset$ in Pérez–Arenas–Gutiérrez, μ_0 in SPARQL 1.1 §18.5, which also names $Z = \{\mu_0\}$ the join identity.

2 The setting as a parameter

Nothing in what follows fixes what $\mathcal{D}$ is. Every definition and both theorems use only three properties: sources hold graphs, query evaluation is monotone in the graph, and the reference answer is evaluation over the union of the sources. What varies between federation and traversal is not the algebra but *how the set of sources comes to be known*, so we make that a parameter.

Definition 1 (Discovery setting). A *discovery setting* for a BGP B is a tuple $W = \langle \mathcal{D}, \text{data}, \sigma, \theta \rangle$ where

- $\mathcal{D}$ is the universe of available sources,
- $\text{data} : \mathcal{D} \rightarrow 2^{\mathcal{T}}$ gives the graph each source holds,
- $\sigma \subseteq \mathcal{D}$ is the set of *seeds*, the starting state, and
- $\theta : \mathcal{D} \times 2^{\mathcal{T}} \rightarrow 2^{\mathcal{D}}$ is the *discovery function*, mapping a retrieved source together with its data to the sources it exposes.

θ may depend on B ; we suppress this in the notation.

Definition 2 (Reachable set, reference answer). Put

$$\mathcal{D}_0 := \sigma, \quad \mathcal{D}_{i+1} := \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \theta(d, \text{data}(d)).$$

The *reachable set* $\text{reach}(W) := \bigcup_{i \geq 0} \mathcal{D}_i$ is the least fixpoint of this monotone operator, and the *reference answer* to B is $\llbracket B \rrbracket_{G_{\text{reach}(W)}}$. Correctness of a delegation is always relative to it.

θ	σ	recovers
$\theta \equiv \emptyset$	all members	classical federated query processing
all URIs occurring in $\mathbf{data}(d)$	seed URIs	LTQP under c_{All}
URIs of triples matching some $tp \in B$	seed URIs	LTQP under c_{Match}
the subweb spec. inside $\mathbf{data}(d)$	the user's document	guided traversal
a link path expression from the query	seed URIs	LDQL

Definition 3 (State at time T , frontier, termination). A *state* at time T is a set $\mathcal{D}'(T)$ with $\sigma \subseteq \mathcal{D}'(T) \subseteq \text{reach}(W)$, non-decreasing in T . Its *frontier* is

$$F(T) := \left(\bigcup_{d \in \mathcal{D}'(T)} \theta(d, \mathbf{data}(d)) \right) \setminus \mathcal{D}'(T),$$

the sources exposed by what has been retrieved but not yet retrieved themselves. *Discovery has terminated* at T if $F(T) = \emptyset$, equivalently $\mathcal{D}'(T) = \text{reach}(W)$.

Remark (Federation is the degenerate case). Classical federation is $\theta \equiv \emptyset$ with σ the configured member list. Then $\text{reach}(W) = \sigma$ and $F(0) = \emptyset$: discovery terminates before the first request. This is exactly why exclusive groups are sound there, and why the impossibility result below has nothing to say about it.

Remark (The reference answer is not the Web). $\llbracket B \rrbracket_{G_{\text{reach}(W)}}$ is evaluation over the union of the *reachable* sources, not over the Web. Taking θ to expose every URI and $\mathcal{D}$ to be every document would give full-web semantics, which is not finitely computable; the traversal instantiations keep $\text{reach}(W)$ bounded by the seeds and the criterion.

Remark (How Δ is obtained differs). In a federated setting Δ comes from cheap probes — ASK requests, a VoID description, an index — independently of retrieving the data. Under traversal there is no such probe: $\Delta(t, d)$ becomes known exactly when $\mathbf{data}(d)$ does, so Δ collapses into the state. The source-description language is therefore strictly poorer in the traversal instantiations, which strengthens rather than weakens the impossibility argument.

Remark (Beyond RDF). Removing triple patterns and dereferencing leaves monotone conjunctive queries over horizontally partitioned data, where a fragment may be shipped whole to one partition exactly when no answer spans partitions. That is the classical localisation condition for distributed query processing; the RDF instance merely supplies particular notions of partition and of link.

3 Relevance and delegation

Notation. For a set of sources $A \subseteq \mathcal{D}$ we write $G_A := \bigcup_{d \in A} \mathbf{data}(d)$ for the union of their graphs. The three instances used below are $G_{\mathcal{D}}$ (the whole federation), $G_{\mathcal{D} \setminus S}$ (the non-delegated sources) and G_A for an arbitrary cover A in the coverage section.

Definition 4 (Source relevance). Let $\mathcal{P}$ be the set of triple patterns and $\mathcal{D}$ the set of all sources. Write $\llbracket t \rrbracket_d$ for the evaluation of t over the data of d . The *relevance indicator* $\Delta : \mathcal{P} \times \mathcal{D} \rightarrow \{0, 1\}$ is

$$\Delta(t, d) = \begin{cases} 1, & \text{if } \llbracket t \rrbracket_d \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} \quad \mathcal{D}_t := \{d \in \mathcal{D} \mid \Delta(t, d) = 1\}.$$

Δ is obtained for the patterns of the query under evaluation only — one ASK probe per pattern per source, as in FedX.

Definition 5 (Federation instance). A *federation instance* is a function $\delta : \mathcal{D} \rightarrow 2^{\mathcal{T}}$ giving the graph held by each source. It is *consistent* with Δ for $t_1, \dots, t_n$, written $\delta \models \Delta$, if

$$\Delta(t_i, d) = 1 \iff \llbracket t_i \rrbracket_{\delta(d)} \neq \emptyset \quad \text{for all } i \in \{1, \dots, n\}, d \in \mathcal{D}.$$

Remark. Δ is what the engine *knows*; δ is what is *there*. Many δ give the same Δ , and the planner must be correct for all of them. Do not call this a “source assignment”: FedQPL Def. 8 already uses that term for a query plan.

Definition 6 (Delegation). Let B be a BGP and $B' \subseteq B$ the delegated fragment. Let $S \subseteq \mathcal{D}$ be a set of sources such that every $d \in S$ provides an interface $I_d = (L_d, \varrho_d)$ with $B' \in L_d$; that is, each delegated source accepts B' as a *single* request, returning $\varrho_d(B', \text{data}(d)) = \llbracket B' \rrbracket_{\text{data}(d)}$. The delegated plan obtains for B'

$$R := \left(\bigcup_{d \in S} \varrho_d(B', \text{data}(d)) \right) \cup \llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}},$$

and the delegation is *correct* if

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}}.$$

Remark (Where the request granularity lives). Semantically $\llbracket B' \rrbracket_{\text{data}(d)} = \bowtie_{t \in B'} \llbracket t \rrbracket_{\text{data}(d)}$, so over a single source one conjunctive request and $|B'|$ pattern requests return the same answers. The two *plans* are distinguished by the order of union and join:

$$\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} \quad \text{versus} \quad \bowtie_{t \in B'} \bigcup_{d \in S} \llbracket t \rrbracket_{\text{data}(d)}.$$

Union outside the join is one request per source; join outside the union is one request per pattern per source. Only the first is delegation.

Remark (Capability and cost). The condition $B' \in L_d$ is what makes “single request” meaningful, and it is the point at which interface heterogeneity enters: a triple-pattern-only interface has $L_d = \mathcal{P}$ and admits no B' with $|B'| > 1$, while a composite resource advertising the path tp_1/tp_2 has $L_d = \{tp_1/tp_2\}$, so $B' \in L_d$ is exactly the requirement that the delegated fragment match what the resource publishes. Counting requests as in FedQPL’s sa-cost, delegation costs $|S|$ where the per-pattern route costs $|B'| \cdot |S|$.

Remark (Soundness is free). $\text{data}(d) \subseteq G_{\mathcal{D}}$ and $G_{\mathcal{D} \setminus S} \subseteq G_{\mathcal{D}}$, so $R \subseteq \llbracket B' \rrbracket_{G_{\mathcal{D}}}$; with $\bowtie$ monotone, the left-hand side is always a subset of the right. Delegation can never produce a wrong answer, only lose results. The whole content of the definition is completeness.

Remark (Sanity checks). $S = \emptyset$ gives $R = \llbracket B' \rrbracket_{G_{\mathcal{D}}}$: not delegating is trivially correct. $S = \mathcal{D}$ gives $R = \bigcup_{d \in \mathcal{D}} \llbracket B' \rrbracket_{\text{data}(d)}$: the no-inter-source-join condition. $S = \{k\}$ with k exclusive gives $R = \llbracket B' \rrbracket_{\text{data}(k)} = \llbracket B' \rrbracket_{G_{\mathcal{D}}}$: FedX’s exclusive group, and FedQPL Def. 7 with $\varphi = \text{req}_k^{B'}$.

Remark (What is lost). Since $\llbracket B \rrbracket_G = \llbracket B' \rrbracket_G \bowtie \llbracket B \setminus B' \rrbracket_G$, the gap between R and $\llbracket B' \rrbracket_{G_{\mathcal{D}}}$ consists exactly of the B' -solutions spanning a boundary introduced by the grouping: across the $S / \mathcal{D} \setminus S$ split, or between two distinct sources inside S . Such a solution is harmless when it has no join partner in $B \setminus B'$, since it would have been discarded anyway.

Remark (Why the two-block form does not work). The variant $\llbracket B \rrbracket_{\mathcal{D}} = \llbracket B \rrbracket_{\mathcal{D} \setminus \{d\}} \cup \llbracket B \rrbracket_d$ forbids only joins crossing the d boundary; it is blind to joins *inside* the block $\mathcal{D} \setminus \{d\}$. Take $\mathcal{D} = \{d, d_1, d_2\}$ and $B = \{t_1, t_2\}$ sharing a variable, with t_1 matching only at d_1 , t_2 only at d_2 , and the two joining. The equation holds, yet an inter-source join exists.

Theorem 1 (Correct delegation). Let $B' = \{t_1, \dots, t_n\}$. Delegation of B' to S is correct for every federation instance consistent with Δ if and only if

$$\left(\exists k \in S : \bigcup_{i=1}^n \mathcal{D}_{t_i} \subseteq \{k\} \right) \quad \text{or} \quad \left(\bigcup_{i=1}^n \mathcal{D}_{t_i} \right) \cap S = \emptyset.$$

Remark. All relevant sources lie inside a single delegated source, or none of them lies in S at all — the latter being the “delegating to an irrelevant source is harmless” case, which plain exclusivity does not capture. Anything in between admits a spanning witness. Both disjuncts are forced by the poverty of Δ : recording only per-pattern relevance, it cannot reveal whether a join actually crosses sources. The quantifier over instances is essential — on a single fixed instance the “only if” fails, since two sources holding identical data are both relevant yet delegating to either is correct.

4 Incomplete discovery

Definition 1 gives Δ over all of $\mathcal{D}$, so an engine holding Δ knows everything. To say anything about discovery we must separate what is true from what has been seen.

Definition 7 (Observation, completion, decision procedure). At time T the engine has retrieved $\mathcal{D}'(T)$ and knows $\text{data}(e)$ for each $e \in \mathcal{D}'(T)$; write $O(T) := (\mathcal{D}'(T), (\text{data}(e))_{e \in \mathcal{D}'(T)})$. It can therefore also compute the frontier $F(T)$, but knows nothing of $\text{data}(d)$ for $d \in F(T)$. A *completion* of $O(T)$ is any discovery setting $\widehat{W} = \langle \widehat{\mathcal{D}}, \widehat{\text{data}}, \sigma, \theta \rangle$ agreeing with $O(T)$ on $\mathcal{D}'(T)$; the adversary’s freedom is exactly the data behind the frontier. A *decision procedure* is a partial function Π from observations to pairs (B', S) with $S \subseteq \mathcal{D}'(T)$, and is *sound* if whenever $\Pi(O(T)) = (B', S)$, delegating B' to S is correct in *every* completion of $O(T)$.

Theorem 2 (No certificate under incomplete discovery). *Let $O(T)$ be an observation at which discovery has not terminated, i.e. $F(T) \neq \emptyset$. Then there are $B', S \subseteq \mathcal{D}'(T)$ and completions $\widehat{\mathcal{D}}_1, \widehat{\mathcal{D}}_2$ of $O(T)$ such that delegating B' to S is correct in $\widehat{\mathcal{D}}_1$ and incorrect in $\widehat{\mathcal{D}}_2$. Consequently no sound decision procedure can output (B', S) .*

Proof. Take $B' = B = \{t_1, t_2\}$ with $t_1 = (?x, p, ?y)$ and $t_2 = (?y, q, ?z)$, so that $B \setminus B' = \emptyset$ and the outer join in Definition 3 is with $\{\mu_\emptyset\}$, i.e. the identity. Let the observation be $\mathcal{D}'(T) = \{e\}$ with $\text{data}(e) = \{(a, p, b)\}$, and let $S = \{e\}$.

World 1. Let $\widehat{\mathcal{D}}_1 = \{e\}$. Here $\mathcal{D}_{t_1} = \{e\}$ and $\mathcal{D}_{t_2} = \emptyset$, so $\llbracket B' \rrbracket_{G_{\widehat{\mathcal{D}}_1}} = \emptyset$, and likewise $R = \emptyset$. The delegation is correct.

World 2. Let $\widehat{\mathcal{D}}_2 = \{e, d^*\}$ with $\text{data}(d^*) = \{(b, q, c)\}$, where $d^* \in F(T)$ sits on the frontier. Now

$$\llbracket B' \rrbracket_{G_{\widehat{\mathcal{D}}_2}} = \{ \{ ?x \mapsto a, ?y \mapsto b, ?z \mapsto c \} \} \neq \emptyset,$$

whereas both terms of R vanish:

$$\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} = \llbracket B' \rrbracket_{\text{data}(e)} = \emptyset, \quad \llbracket B' \rrbracket_{G_{\widehat{\mathcal{D}}_2 \setminus S}} = \llbracket B' \rrbracket_{\text{data}(d^*)} = \emptyset,$$

since e has no t_2 -match and d^* has no t_1 -match. Hence $R = \emptyset$ and the answer is lost: the delegation is incorrect.

Both completions agree with $O(T)$ on $\mathcal{D}'(T)$, so no function of $O(T)$ can distinguish them. $\square$

Remark (Why Δ alone cannot establish this). The witness must exhibit concrete triples. At the level of Δ the two worlds are distinguished only by $\Delta(t_2, d^*) = 1$, and that alone does not settle correctness: if d^* merely *duplicated* data already held by e , no new solution would appear and the delegation would remain correct. What breaks it is a triple that is new *and* joins across the boundary.

Remark (Relevance elsewhere is not by itself a problem). Discovering $\Delta(t, d) = 1$ for an undiscovered $d \notin S$ does *not* break the delegation. Since $d \in \mathcal{D} \setminus S$, the second term $\llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}}$ of R still collects everything d can answer on its own. Only solutions *spanning* S and its complement — or spanning two distinct sources inside S — are lost. This is the narrower fact the impossibility argument needs.

Remark (Side conditions). The construction needs B' connected (two patterns sharing a variable), $S \neq \emptyset$, and at least one pattern of B' matched within S ; otherwise there is nothing for the adversary's triple to join to. Taking $B' = B$ removes any further condition, since then no join partner in $B \setminus B'$ is required.

Remark (What this asks of the assumption). The argument turns entirely on the adversary being free to place arbitrary data behind an unfetched URI. Any fix must remove that freedom rather than improve the source descriptions: no refinement of Δ over $\mathcal{D}'(T)$ helps, because both worlds agree there. This is what coverage and authority declarations supply in the next section.

5 Coverage

Definition 8 (Coverage). $\text{cov} : \mathcal{D} \rightarrow 2^{\mathcal{D}}$, in two flavours:

$$d \in \text{cov}_{\text{agg}}(c) \iff \text{data}(d) \subseteq \text{data}(c), \quad d \in \text{cov}_{\text{idx}}(x) \iff x \text{ publishes } \Delta(\cdot, d).$$

An aggregator lets you *act*; an index lets you *decide*. Lift to sets by $\text{cov}(A) := \bigcup_{c \in A} \text{cov}(c)$, and call A a *cover* of $\mathcal{D}$ if $\text{cov}(A) = \mathcal{D}$.

Remark. cov is extensive, monotone and idempotent (coverage is transitive), hence a closure operator on $2^{\mathcal{D}}$. Covers form an up-set; the minimal ones are the objects of interest.

Lemma 1 (Answer equivalence). *If A is an aggregator-cover of $\mathcal{D}$ then $G_A = G_{\mathcal{D}}$, and therefore $\llbracket Q \rrbracket_{G_A} = \llbracket Q \rrbracket_{G_{\mathcal{D}}}$ for every monotone Q .*

Lemma 2 (Observation asymmetry). *Let $\mathcal{D}'_t(T) := \{d \in \mathcal{D}'(T) \mid \Delta(t, d) = 1\} \subseteq \mathcal{D}_t$. Then*

$$|\mathcal{D}'_t(T)| \geq 2 \implies |\mathcal{D}_t| \geq 2, \quad \text{but} \quad |\mathcal{D}'_t(T)| = 1 \not\Rightarrow |\mathcal{D}_t| = 1.$$

Non-exclusivity is soundly concludable from a finite observation; exclusivity is not. That asymmetry is the impossibility result.

Definition 9 (Discovery closure). Recall $\theta(d, \text{data}(d))$ from Definition 1. A set A is *discovery-closed* if

$$\forall c \in A : \theta(c, \text{data}(c)) \subseteq \text{cov}(A).$$

Lemma 3 (Frontier). *If A is discovery-closed and $\text{seeds} \subseteq A$, then $\text{reach}(\text{seeds}) \subseteq \text{cov}(A)$.*

This converts the unverifiable global condition “I have seen everything” into the local, checkable one “my frontier is covered”.

Theorem 3 (Decidability under coverage). *If at time T some $A \subseteq \mathcal{D}'(T)$ is discovery-closed with admissible index coverage, then $\mathcal{D}_t$ is computable from $\mathcal{D}'(T)$ alone, and exclusivity of k for $\{t_1, \dots, t_n\}$ is decidable at time T .*

Theorem 4 (Impossibility without coverage). *If $\text{cov}(\mathcal{D}'(T)) \subsetneq \mathcal{D}$, then for every k with $k \in \mathcal{D}_{t_i}$ for all i there is a completion of $\mathcal{D}$ consistent with every observation made up to T under which delegating $\{t_1, \dots, t_n\}$ to k is incorrect.*

Corollary 1. *Without coverage declarations, exclusivity is eventually computable but not finitely computable (Hartig & Freytag, Defs. 13–14). With an admissible discovery-closed index cover it becomes finitely computable.*

Remark. Choosing the cheapest $A \subseteq \mathcal{D}'(T)$ with $\text{cov}(A) = \mathcal{D}$ is weighted Set Cover: NP-hard, greedy gives $H_n \approx \ln n$. FedQPL already proves the analogous hardness for minimal source assignments by reduction from node cover (Def. 10 and Prop. 2), so that precedent is citable rather than new.

Remark. Coverage and the namespace scope claim are one mechanism: a claim over N covers exactly the sources whose data lies in S_N . Coverage is the extensional presentation, the scope claim the intensional one.